Warnings on Top: Risk Language and the Rise of Artificial-Intelligence Disclosure in 10-K Filings, 2015 to 2026

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Abstract

The share of Form 10-K filings containing the phrase "artificial intelligence" rose from 0.5 percent in 2015 to 49.2 percent in 2025. I classify all 2,242 AI sentences in 315 sampled filings by what they say and whether they assert that the filer itself uses AI. Capability claims per AI-mentioning filing roughly doubled, from 1.8 to 3.8, but risk-factor sentences rose from 0.3 to 6.4 and account for 63 percent of the added volume, so the risk share of a filing's AI language went from 12 to 63 percent. The share of AI-mentioning filings with a substantive own-use statement fell from 60 to 51 percent, an imprecise decline, and conditional on how much AI text a filing contains, recent filers are much less likely to assert use. The ratio of mention counts to substantive own-use assertions is about 1.7 before 2020 and 2.0 since 2024. Sentences were labelled by blinded language-model raters, validated against known answers and a blinded model review but not yet against human coders, for whom a validation sample is published. Language models also wrote the code and drafted this paper under the author's direction, as the AI-use statement describes.

Keywords: artificial intelligence, corporate disclosure, 10-K, risk factors, textual analysis, boilerplate, AI washing.

1. Introduction

Counting the annual reports that mention artificial intelligence is the cheapest available indicator of how far AI has spread through public companies. The filings are public, the phrase is unambiguous, and EDGAR's full-text search returns the count in a second. The count has risen in a way that looks like a diffusion curve: fewer than one 10-K in two hundred contained the phrase in 2015, one in two did in 2025, and three in five have so far in 2026 (Table 1). Published adoption measures built from corporate text, such as the AI-disclosure measures of Barrios et al. (2025) and the talk-versus-walk comparison of Li (2025), are more careful than a raw count, but they share its premise that what firms write about AI is informative about what they do with it.

This paper asks what the rising count is made of. A filing can mention AI to describe its own product, to warn that AI is a risk, to say competitors are using it, to describe board oversight, or in passing. Only the first, and some risk sentences that mention the filer's use along the way, say anything about adoption. If the growth comes from the other kinds of sentence, the count misstates adoption, and it misstates it differently in different years.

I draw a random sample of AI-mentioning 10-Ks for every year from 2015 to 2026, extract every AI sentence, and classify each one by what it says and by whether it asserts that the filer itself uses AI.

Three findings follow.

First, the growth in AI language is mostly risk language added on top of product language rather than replacing it. Capability claims per AI-mentioning filing roughly doubled between 2015 to 2019 and 2024 to 2026, while risk-factor sentences rose nineteen-fold and supplied 63 percent of the added volume. By 2024 to 2026 nearly half of all AI sentences sit in Item 1A, the risk-factor section, against 14 percent before 2020.

Second, the filers who mention AI are no more likely to state their own use of it than before, and on a stricter definition they are less likely. About three in five AI-mentioning filings contain some own-use sentence in every era, but the late-era figure leans on risk sentences that mention use in passing. Counting only substantive own-use statements, the share falls from 60 percent before 2020 to 51 percent since 2024, and holding constant how much AI text a filing contains, a recent filing is far less likely to assert use than an earlier one with the same amount.

Third, verbatim boilerplate is rare but concentrated. Of capability sentences, none is shared word for word with five or more other registrants, against 18 percent of risk sentences, and the share of a filing's AI language that is copied across filers rose from zero before 2020 to 12 percent since 2024.

The study extends a pilot on 60 filings from 2025 (Shah 2026). Its hypotheses and sampling plan were preregistered before any filing was drawn, and every sentence, label, prompt and line of code is public. Section 7 lists where the analysis departs from that plan.

2. Related work

Textual analysis of filings has long warned that a word count built for one purpose misleads when used for another (Loughran and McDonald 2011, 2016, Gentzkow, Kelly and Taddy 2019). Several strands bear directly on this paper.

Length, boilerplate and stickiness. Li (2008) linked annual-report readability to earnings persistence. Dyer, Lang and Stice-Lawrence (2017) showed that 10-K length, boilerplate, stickiness and redundancy rose from 1996 to 2013, with risk factors among the few topics that account for most of the growth, and Cazier and Pfeiffer (2016) decomposed 10-K length. Lang and Stice-Lawrence (2015) measured boilerplate from phrases common across filings. Brown and Tucker (2011) and Cohen, Malloy and Nguyen (2020) studied year-over-year change in filing text and showed that revisions are rare and informative. Hanley and Hoberg (2010) separated standard from informative content in IPO prospectuses, the closest precedent for the distinction between template and substance that this paper draws for AI language.

Risk factors. Mandated risk factors carry firm-specific information on average (Campbell et al. 2014, Kravet and Muslu 2013), but generic risk language is discounted relative to specific language (Hope, Hu and Lu 2016). Redigolo and Marcarini (2026) find that Item 1A text became more standardised after ChatGPT. Two regulatory changes matter for interpreting any post-2020 shift in risk language: the 2020 amendments to Regulation S-K, which discouraged generic risk factors and required them to be grouped under a "General Risk Factors" caption (U.S. Securities and Exchange Commission 2020), and the 2023 cybersecurity rule, which added Item 1C for fiscal years ending on or after 15 December 2023 (U.S. Securities and Exchange Commission 2023).

Text-based firm measures and AI. Hassan et al. (2019) and Florackis et al. (2023) built firm-level risk exposures from text and validated them against outcomes, the standard a text measure should meet. For AI specifically, Babina et al. (2024) measure investment from employee resumes, Eisfeldt,

Schubert and Zhang (2023) from occupational exposure, and Kalyani et al. (2025) trace how new technologies diffuse across firms. Li (2025) finds that AI talk on earnings calls does not predict AI hiring, while Barrios et al. (2025) find that AI disclosures in 10-Ks and calls do predict later efficiency and patents. Juma'h (2026) documents vague and promotional AI language in 10-Ks, and Zhang and Rivera (2026) and Ofir (2026) treat AI washing as a governance and a legal problem. Census Bureau surveys measure AI use directly (Bonney et al. 2024, McElheran et al. 2024). This paper does not propose a new adoption measure. It decomposes the raw count that these measures refine.

Language models as annotators. Generative models can match or beat crowd coders on annotation (Gilardi, Alizadeh and Kubli 2023), and de Kok (2025) gives a framework for their use in accounting research. Their labels contain errors that bias downstream estimates unless corrected with a human-coded subsample (Egami et al. 2023, Angelopoulos et al. 2023). That correction is the main piece of work this paper has not yet done, and Section 6 says what it requires.

3. Data

3.1 The frame and the denominator

For each calendar year from 2015 to 2026 the frame is every original Form 10-K (not an amendment) whose main document contains the exact phrase "artificial intelligence" in EDGAR full-text search, counted by filing year and running through 18 October 2026. Exhibit-only matches are excluded. The denominator is every original 10-K in the EDGAR full index for the year. Across the twelve years the frame holds 13,586 of 82,684 10-Ks. Years are years of filing, so 2026 contains real annual reports already on EDGAR, mostly for fiscal 2025. Because most 10-Ks are filed in the first quarter, 2026 is nearly complete, and its share will move slightly by December.

year	frame	all 10-Ks	raw share
2015	38	7,659	0.50%
2016	45	7,259	0.62%
2017	110	6,984	1.58%
2018	271	6,827	3.97%
2019	400	6,660	6.01%
2020	511	6,508	7.85%
2021	734	6,933	10.59%
2022	1,071	7,538	14.21%
2023	1,206	7,118	16.94%
2024	2,324	6,631	35.05%
2025	3,162	6,431	49.17%
2026 to 18 Oct	3,714	6,136	60.53%

Table 1. Original 10-K filings whose main document contains "artificial intelligence", all original 10-K filings, and the raw mention share, by calendar year of filing.

The frame is defined by one phrase. A filing that writes "machine learning" or "AI" but never "artificial intelligence" is outside it, so every adoption share below is a share of filings that use the phrase, and a lower bound on the share of all 10-Ks that discuss AI in any words.

3.2 The sample and the weights

Within each year the frame was collapsed to one filing per registrant and a random sample was drawn under a fixed seed: 30 filings in each of 2015 to 2017 and 25 in each year from 2018 to 2026, 315 filings from 281 registrants. The draw is far from proportional. It takes 30 of 36 registrants in 2015 and 25 of 3,710 in 2026, so every estimate that pools years weights each sampled filing by its year's registrant count divided by the number drawn, and the bootstrap resamples filings within year. The eras are therefore dominated by their later years: 2018 and 2019 carry 78 percent of the weight of 2015 to 2019, and 2025 and 2026 carry 75 percent of 2024 to 2026. Weighted estimates describe the population of AI-mentioning filings, not the average year. Unweighted estimates are published beside them in the repository.

3.3 Extraction

Each main document was stripped of markup and split into sentences by a fixed splitter. A sentence entered the corpus if it matched the AI term list, was 60 to 1,200 characters long, and was under a quarter digits. All 315 filings yielded 2,590 candidate sentences. Four sampled registrants carry the letters AI in their own names, one of which, C3.ai in 2023, contributed 448 candidates. A name screen and a blinded read removed 348 sentences that mention AI only through a registrant's name, leaving 2,242 sentences in 313 filings, a median of 3 per filing. Each sentence was also tagged with the 10-K item it sits in, from the item headings in the document.

4. Method

4.1 What each sentence is judged on

Each sentence receives a class and an own-use judgement.

The **class** is one of five. `capability` means the sentence says the filer's own product, service or operation uses AI and attributes a capability or benefit to it. `risk` is risk-factor language about AI. `market` describes AI in the industry or at competitors. `governance` describes policy or oversight of AI. `other` covers definitions, cross-references and everything else, including a sentence that says the filer uses AI without saying what the AI achieves.

Own use is true when the sentence states that the filer itself currently uses, offers, develops or is implementing AI, whether or not a benefit is attributed, and false for hypothetical use, statements about others, definitions and risk language that does not assert actual use. A risk sentence that asserts use in passing, "our use of AI in underwriting exposes us to model risk", is true.

Because own use can be carried by a risk sentence, the paper reports two filing-level indicators. A filing has **any own-use statement** if at least one of its sentences is coded own use. It has a **substantive own-use statement** if at least one own-use sentence is not risk-factor language and is not shared verbatim with five or more other registrants. The second definition was added after the referee review described in Section 7 and is exploratory.

4.2 The estimands

For year t, let m_t be the share of all 10-Ks in the frame, and let p_t be the design-weighted share of frame filings with a substantive own-use statement. The share of all 10-Ks that make such a statement is $a_t = m_t\, p_t$, and the **mention-to-assertion ratio** is $m_t/a_t = 1/p_t$, the factor by

which the raw count exceeds the count of filings asserting use. The ratio is reported for both own-use definitions. It measures statements in filings, not adoption. A filer can use AI without saying so, and a statement is not verified.

Composition is reported two ways: the mean number of AI sentences per filing in each class, which shows what was added, and the share of a filing's AI sentences in each class, averaged over filings, which shows the resulting mix. Both are design-weighted.

4.3 Raters

The sentences were labelled by language models, not people. Each sentence was shown, with the registrant's name and ticker redacted and in an order shuffled across filings and years, to two of six independent instances of the same commercial language model, each with no shared context and the literal prompt committed in `docs/RATER_PROMPT.md`. The 155 sentences on which the two passes disagreed were rated a third time, blind, by a larger model from the same family, mixed with 155 agreed sentences. The label used throughout is the majority of three. Thirty sentences with known answers from the pilot's calibration set were inserted unmarked. Their answers had been fixed by a blinded model review in the pilot, not by a human, so they test consistency with the pilot rather than correctness.

A seeded sample of 120 sentences, stratified by class and era, with ten duplicates, was then re-read blind by the largest model in the same family, which also orchestrated the study. Its verdicts give a survival rate for each class and each own-use value, which is used to correct the composition shares and, by simulation, the filing-level fractions.

All raters, the third pass and the calibration reviewer come from one model family. Their errors may be correlated, and agreement among them measures consistency, not validity. Section 6 describes the human validation that addresses this.

4.4 Phrase reuse

For each sentence, the twelve consecutive words starting at the first AI term were searched as an exact phrase across all 10-K filings since 2001, and the number of other registrants whose filings contain the phrase was recorded. A sentence is flagged as boilerplate when five or more other registrants have used its phrase. The window was fixed by a pilot before any corpus sentence was searched, and an eight-word run is reported as a check. The search covers all years, so a flagged sentence may have been copied from a template or may have become one later.

4.5 Statistics

Intervals are 95 percent percentile bootstrap intervals from 10,000 replicates that resample filings within year, except where noted. Regressions are logistic, weighted by the design weights with each filing counted equally within year, and cluster standard errors by registrant.

5. Results

5.1 The labels are consistent with each other

The two blind passes gave the same class on 94.6 percent of sentences (Krippendorff's alpha 0.924) and the same own-use judgement on 97.7 percent (alpha 0.952). Of the 121 class disagreements,

73 were capability against other. All 60 judgements on the 30 known-answer sentences matched. In the blinded calibration, 90 percent of majority labels survived: risk 96 percent, other 96, market 89, capability 87 and governance 5 of 8. All four overturned capability labels went to other. Own-use labels survived at 94 percent when true and 89 percent when false. These are model-on-model checks and are a floor, not a validation.

5.2 Risk language was added on top

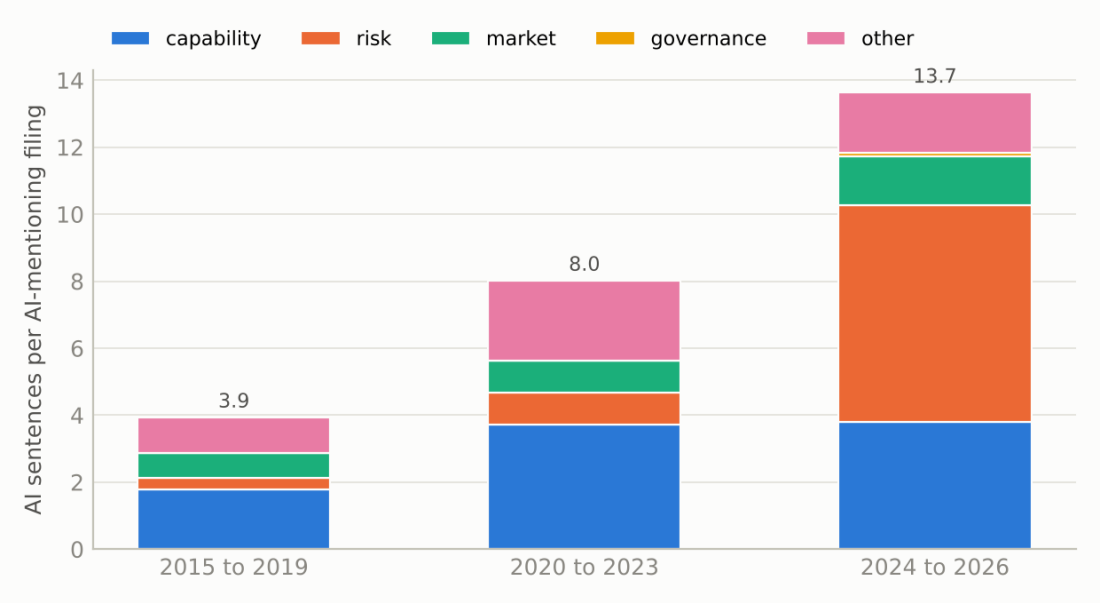


Figure 1: Mean AI sentences per AI-mentioning filing, by class and era. Majority-of-three labels, design-weighted. Totals above the bars.

Figure 1 and Table 2 show what an AI-mentioning filing contains. In 2015 to 2019 it held 3.9 AI sentences, 1.8 of them capability claims and 0.3 risk sentences. In 2024 to 2026 it holds 13.7, of which 3.8 are capability claims and 6.4 are risk sentences. Capability claims per filing roughly doubled, and almost all of that happened by 2020 to 2023. Risk sentences rose nineteen-fold, almost all after 2023. Of the 9.7 sentences added between the first and last era, 6.1 are risk (63 percent), 2.0 capability (21 percent), 0.7 market and 0.7 other.

mean per filing	2015 to 2019	2020 to 2023	2024 to 2026
capability	1.77 [1.21, 2.41]	3.72 [2.20, 6.08]	3.81 [2.14, 5.78]
risk	0.35 [0.14, 0.66]	0.97 [0.52, 1.49]	6.44 [4.49, 8.79]
market	0.75 [0.47, 1.08]	0.96 [0.43, 1.77]	1.46 [0.59, 2.77]
governance	0.00	0.00	0.11 [0.04, 0.20]
other	1.06 [0.58, 1.75]	2.37 [0.94, 4.29]	1.79 [1.12, 2.55]
all AI sentences	3.94 [2.82, 5.57]	8.03 [4.71, 13.16]	13.66 [9.33, 18.61]
share that is risk, percent	12.0 [5.7, 19.6]	18.4 [12.0, 25.6]	63.0 [55.4, 70.4]
share that is capability, percent	40.6 [31.3, 50.2]	43.1 [35.5, 50.9]	15.7 [10.8, 21.1]
filings, sentences	139, 490	99, 735	75, 1,012

Table 2. AI sentences per AI-mentioning filing by class, and the share of a filing's AI sentences in the two main classes, averaged over filings. Design-weighted, majority-of-three labels, 95 percent bootstrap intervals. The 2020 to 2023 means include the C3.ai filing, with 154 sentences, which widens their intervals.

The share of a filing's AI language that is risk therefore rose from 12 to 63 percent and the capability share fell from 41 to 16 percent, but the fall in the capability share is dilution, not substitution. A reader who sees only the shares would conclude that firms stopped describing their AI. They did not. They kept describing it and added warnings.

The shift is a step, not a trend. In a sentence-level regression, the odds that an AI sentence is risk language are 12.5 times higher in filings made from 2024 onward than in 2015 to 2019 (interval 5.8 to 27.3), while 2020 to 2023 does not differ detectably from the first era. Calibration correction does not change this: the corrected rise in the risk share is 49.8 points (39.0 to 60.1).

5.3 Where the sentences sit

The item tags locate the change. Before 2020, 64 percent of AI sentences were in Item 1, the business description, and 14 percent in Item 1A, risk factors. In 2024 to 2026, 40 percent are in Item 1 and 47 percent in Item 1A, and 83 percent of the AI sentences in Item 1A are risk-class. The new cybersecurity item, Item 1C, holds 0.5 percent of late-era AI sentences, so the 2023 cybersecurity rule is not where the AI language went. The 2020 Regulation S-K amendments cannot explain the timing either, since the risk share in 2020 to 2023 is close to its pre-2020 level and the jump comes in 2024, with the first annual reports written after the release of ChatGPT. The design cannot separate that release from other changes in 2023, including the Commission's growing attention to AI claims (U.S. Securities and Exchange Commission 2024, 2025).

5.4 Fewer filers assert use than the count implies

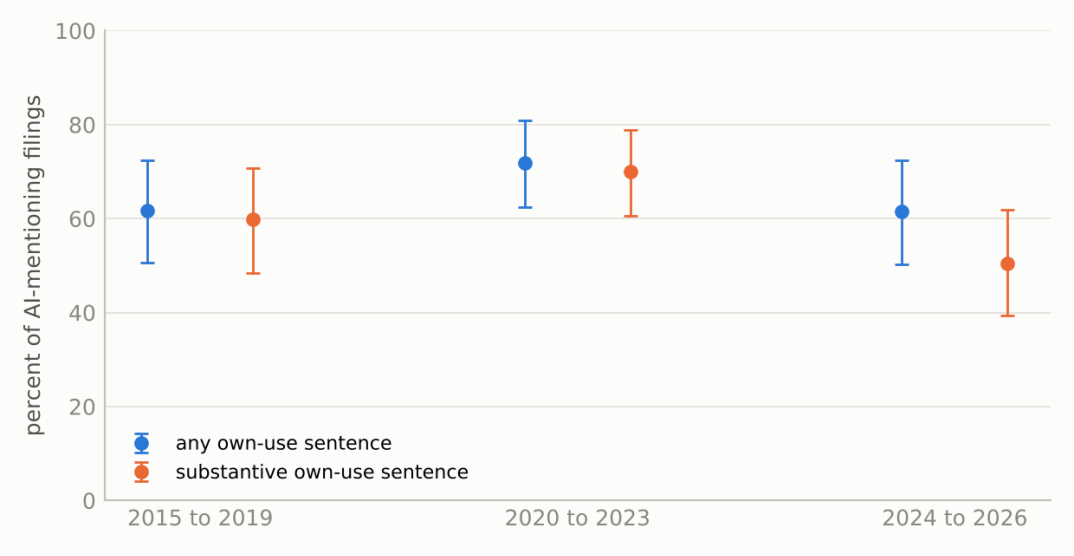


Figure 2: Share of AI-mentioning filings with any own-use sentence and with a substantive own-use sentence, by era, with 95 percent bootstrap intervals. Design-weighted.

Table 3 and Figure 2 give the filing-level results. The share of AI-mentioning filings with any own-use sentence is 61.6, 71.9 and 61.5 percent across the three eras, with no trend. That stability is partly made of risk language: the share of filings whose only own-use sentences are risk sentences rose from 1.9 percent before 2020 to 11.0 percent since 2024. On the substantive definition the share is 59.8, 70.0 and 50.5 percent, a late-minus-early fall of 9.3 points whose interval, from 24.8 down to 6.5 up, includes zero.

	2015 to 2019	2020 to 2023	2024 to 2026
any own-use statement, percent	61.6 [50.5, 72.5]	71.9 [62.5, 80.9]	61.5 [50.3, 72.4]
substantive own-use statement, percent	59.8 [48.3, 70.8]	70.0 [60.6, 78.9]	50.5 [39.4, 61.8]
own use only in risk sentences, percent	1.9 [0.0, 5.6]	2.0 [0.0, 5.3]	11.0 [4.2, 18.8]
mention-to-assertion ratio, any	1.62 [1.38, 1.98]	1.39 [1.24, 1.60]	1.63 [1.38, 1.99]
mention-to-assertion ratio, substantive	1.67 [1.41, 2.07]	1.43 [1.27, 1.65]	1.98 [1.62, 2.54]

Table 3. Filing-level own-use statements among AI-mentioning filings, design-weighted, with 95 percent bootstrap intervals, and the mention-to-assertion ratio of Section 4.2. Ratio intervals are the reciprocals of the fraction intervals.

The flat headline fraction also hides a length effect. A filing with more AI sentences has more chances to contain an own-use sentence, and late-era filings have three times as many. Holding the number of AI sentences constant, the odds that a 2024 to 2026 filing contains any own-use sentence are 0.21 times those of a 2015 to 2019 filing (interval 0.07 to 0.58). Among filings with one or two AI sentences, 42 percent made an own-use statement before 2020 and 14 percent since 2024. Among those with three to five, the figures are 81 and 52 percent. The proportion of AI sentences that state own use fell from 60 to 50 percent. A recent filing that mentions AI briefly is now most likely to be warning about it.

Rater error matters here because one false own-use label is enough to count a filing. Simulating the calibration error rates through to the filing level, with their uncertainty, moves the any-own-use fractions up to 67.9, 75.7 and 70.9 percent. Noise of that size raises levels by six to nine points in every era and is larger where filings are longer, so it biases toward finding no decline, not toward finding one.

5.5 The mention-to-assertion ratio by year

year	filings	raw share	any own use	substantive	asserting use, share of all 10-Ks	ratio
2015	29	0.50	79.3	79.3	0.4 [0.3, 0.5]	1.26
2016	30	0.62	73.3	73.3	0.5 [0.4, 0.5]	1.36
2017	30	1.58	56.7	56.7	0.9 [0.6, 1.2]	1.76
2018	25	3.97	56.0	56.0	2.2 [1.4, 3.0]	1.79
2019	25	6.01	64.0	60.0	3.8 [2.6, 5.0]	1.56

year	filings	raw share	any own use	substantive	asserting use, share of all 10-Ks	ratio
2020	24	7.85	66.7	62.5	5.2 [3.8, 6.6]	1.50
2021	25	10.59	76.0	76.0	8.0 [6.4, 9.7]	1.32
2022	25	14.21	76.0	76.0	10.8 [8.5, 13.1]	1.32
2023	25	16.94	68.0	64.0	11.5 [8.1, 14.2]	1.47
2024	25	35.05	64.0	56.0	22.4 [15.4, 28.0]	1.56
2025	25	49.17	52.0	40.0	25.6 [15.7, 35.4]	1.92
2026	25	60.53	68.0	56.0	41.2 [29.1, 50.8]	1.47

Table 4. By year: filings drawn, the raw mention share of all 10-Ks, the percent of sampled AI-mentioning filings with any and with a substantive own-use statement, the estimated share of all 10-Ks with any own-use statement, with its bootstrap interval, and the ratio of the raw share to it. Shares of all 10-Ks are percentages.

With 25 filings a year, yearly fractions carry Wilson intervals 28 to 37 points wide, and the yearly ratio ranges from 1.26 to 1.92. The eras are the usable summary. A single year's corrected share, such as the 25.6 percent of 2025 10-Ks estimated to contain an own-use statement, should not be read on its own.

5.6 Verbatim reuse is rare and is risk language

Of the 2,166 sentences whose phrases the search returned, 84.8 percent share their twelve-word phrase with no other registrant. None of 864 capability sentences is shared with five or more others, against 104 of 565 risk sentences (18.4 percent). The flag identifies risk language with 96 percent precision and 18 percent recall, both measured against the model labels. The flagged share of a filing's AI sentences was 0.0 percent before 2020, 2.8 percent in 2020 to 2023 and 12.3 percent since 2024 (interval 7.4 to 17.7). The most widely copied sentences are generic risk factors: the inability to use generative AI (87 other registrants), deficient or biased AI output (78), competitors adopting AI faster (74), AI ethics controversy (68) and the cost of AI regulation (63). The eight-word window raises the flagged share only from 5.0 to 6.2 percent.

Firms also repeat themselves. For the 34 pairs of filings by the same registrant drawn in two years, a median of one year apart, 27 percent of the later filing's AI sentences reuse at least half of their eight-word sequences from the earlier filing's AI sentences.

So AI language is mostly not boilerplate in the verbatim sense that Lang and Stice-Lawrence (2015) measure, and the generic risk factors that are copied are a small part of the risk language that was added. Most of the new risk text is written for the filing, even when it says much the same thing as other filings.

5.7 Sector, size and replication

Pooling years with the design weights, 85.7 percent of technology registrants' AI-mentioning filings state own use (interval 76.3 to 94.0), against 39.5 percent in finance and real estate, 69.8 percent in healthcare and 63.2 percent elsewhere. Sectors are SIC ranges: technology is 3570 to 3579, 3660 to 3679 and 7370 to 7379, healthcare is 2833 to 2836, 3826 to 3851, 5912 and 8000 to 8099, and finance and real estate is 6000 to 6799. Technology registrants fell from 42 percent of the frame

before 2020 to 16 percent since 2024, which contributes to the late-era decline in own use. Larger registrants, above about one billion dollars in assets, devote 57.7 percent of their AI sentences to risk against 34.5 percent at smaller ones, partly because smaller reporting companies need not provide Item 1A.

For 2025, the 25 filings here give a sentence-weighted risk share of 49.5 percent (37.6 to 64.8) against the pilot's 65.5 percent, just outside the interval, and a capability share of 28.5 percent (11.0 to 42.5) against 13.5. On the filing-weighted basis used in this paper the 2025 figures are 67.5 and 11.2 percent. The two studies agree only loosely, as 25 filings would lead one to expect.

5.8 Preregistered predictions

Seven predictions were registered. Each is classified as held (point estimate in the predicted direction, interval excluding the null), held on the point estimate only, or failed on the point estimate.

1. The risk share is higher in 2024 to 2026 than in 2015 to 2019: held, +51.1 points [40.5, 60.8].
2. The own-use share of a filing's AI sentences is lower: held, −20.3 points [−31.9, −8.3]. Weighting sentences equally, the interval includes zero.
3. Fewer than 60 percent of late-era AI-mentioning filings contain any own-use sentence: failed on the point estimate, 61.5 percent [50.4, 72.2]. On the substantive definition the figure is 50.5 percent, but that definition was not registered.
4. Risk sentences are reused at least three times as much as capability sentences by median, and flagged at least twice as often: held on the second criterion (18.4 against 0 percent). The first cannot be evaluated because both medians are zero.
5. The flagged share rises: held, +12.3 points [7.4, 17.7].
6. The mention-to-assertion ratio is larger in the last era: held on the point estimate only, 1.63 against 1.62, interval on the difference [−0.43, 0.43]. On the substantive definition the ratio rises to 1.98, but again that was not registered.
7. Technology registrants have a higher own-use rate: held, +26.9 points [13.8, 39.4].

6. Limitations and what would fix them

No human has validated the labels. Every check in Section 5.1 is a model checking models from the same family, and the calibration reviewer orchestrated the study. A blinded sample of 300 sentences, stratified to oversample the capability-other and risk-own-use boundaries, is published with a coding guide in `data/human_validation/` and `docs/HUMAN_CODING_GUIDE.md`, together with a script that reports human-human agreement, model-human agreement and per-class precision and recall. Until two human coders complete it, the levels in this paper should be read as model estimates. The natural next step after that is to correct the estimates with the human subsample (Egami et al. 2023, Angelopoulos et al. 2023) and to cross-check a subsample with a model from a different developer.

The sample is thin. Twenty-five filings a year support era comparisons, not single-year estimates. Extending the draw to at least 100 filings a year from 2022 is the cheapest improvement to every interval in Tables 3 and 4.

The frame is one phrase. Filings that discuss AI only as "machine learning" or "AI" are excluded. EDGAR returns 2,050 10-K hits for "machine learning" in 2026 against 3,883 for "artificial intelligence", and the overlap was not measured.

Statements are not adoption. The paper measures what filings say. Whether the statements are true, and whether the added risk language carries information as Campbell et al. (2014) found for risk factors generally or is discounted as Hope, Hu and Lu (2016) found for generic ones, needs outcome data this design does not have.

Timing is not identified. The 2024 step coincides with the first post-ChatGPT annual reports, with growing regulatory attention to AI claims, and with the cybersecurity rule. The item tags rule out Item 1C as the channel, but the design cannot separate the others.

Reuse is measured across all years. A flagged sentence may be a template or may have become one, and 3.5 percent of phrases failed to find their own filing, a normalisation failure that biases reuse counts down.

Blinding leaked slightly. Period cues such as "generative AI" appear in 6.7 percent of sentences and a name fragment survived redaction in 3.7 percent. Agreement was not higher on them, but whether the cues shifted labels toward risk was not tested.

7. Departures from the preregistration

The preregistration, sampling, ratings and first analysis were all completed on 18 October 2026, and several choices that shape the reported magnitudes were made after the ratings were seen. Table 5 lists them.

change	when	effect
filings rather than sentences weighted equally in composition shares	after ratings	larger risk shift (51 against 38 points sentence-weighted); both reported
design weights for pooled and era estimates	after the first draft, on an independent check	narrows Prediction 6 from 1.63 against 1.51 unweighted to 1.63 against 1.62
batches of 50 rather than about 40 sentences	during rating	none expected
per-class calibration correction applied to filing-level fractions by simulation rather than analytically	second revision	raises levels six to nine points, no change in direction
mean counts per filing, substantive own use, length-adjusted model, item tags, within-firm reuse, post-2023 regression	second revision, after a referee-style review	exploratory, reframes the headline from substitution to addition
sector ordering under the raw measure and a sector comparison with Census data	registered, not done	needs the sector of every 10-K filer

Table 5. Departures from the preregistered plan. The dated amendments are in `docs/PREREGISTRATION.md`.

8. Conclusion

The number of 10-Ks that mention artificial intelligence grew a hundredfold in a decade, and most of the language that came with it after 2023 is risk-factor text. That text did not replace what firms say about their own AI. Capability claims per filing doubled and then held steady while warnings were stacked on top, until nearly half of all AI sentences sat in Item 1A. The share of AI-mentioning filers that make a substantive statement about their own use, meanwhile, has not grown and may have fallen, and a recent filing with little AI text is now most often a filing that only warns.

For anyone using the count, the practical rule is that the raw number of filings mentioning AI exceeds the number substantively asserting use by a factor of about 1.4 to 1.7 before 2024 and about 2.0 since, with intervals that make any single year unreliable, and that the volume of AI text has become mainly a measure of risk-factor drafting. Both statements rest on model labels that still await the human validation the paper publishes the materials for.

AI-use statement

Language models were used throughout this project. In the measurement, they served as the sentence raters, the third-pass rater, the term-audit reader and the calibration reviewer, as Section 4.3 describes. Beyond the measurement, an AI agent working under the author's direction proposed the research design, wrote the data-collection and analysis code, drafted the preregistration and this paper, searched the literature and checked each reference against its DOI record, and an independent AI reviewer critiqued the draft. The author selected the research question, directed the work, and takes responsibility for the content. The models and versions used for each role are recorded and available from the author.

Data and code

Data, code, the preregistration and its amendments, the rater prompt, every sentence and label, and the human-validation materials are at `https://github.com/ishaanshah101/ai-disclosure-trend-audit`. `src/analyze.py` and `src/revision2.py` produce every number in the paper, and `src/verify_paper.py` checks the numbers quoted here against their output.

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